Mathematics-I | Unit 1

Determinant and Function — GTU Diploma Engineering (DI01000021)

Weightage: 23% (highest-weighted unit in this subject). 9 teaching hours. Covers determinants up to 3rd order, functions and domain, logarithm as a function, and the laws of logarithm.

1.1 Determinant and its Value up to 3rd Order

A determinant is a single number computed from a square matrix. Used for solving simultaneous equations and later for vector cross products in Unit 3. No properties in this syllabus — computation only.

2×2 Determinant

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

Value = $a_1b_2 - a_2b_1$ (cross-multiply diagonally, subtract)

Example 1

$$\begin{vmatrix} 3 & 4 \\ 2 & 5 \end{vmatrix}$$

$= (3)(5) - (2)(4) = 15 - 8 =$ **7**

3×3 Determinant — expand along the first row

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Sign pattern for a 3×3 expansion is fixed: **+ − +**. Memorize it, don't guess it. To get the 2×2 sub-determinant for a given element: delete that element's row and column — nothing else.

Example 2

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$$

$= 1(1\times0 - 4\times6) - 2(0\times0 - 4\times5) + 3(0\times6 - 1\times5)$

$= 1(-24) - 2(-20) + 3(-5) = -24 + 40 - 15 =$ **1**

Where marks are lost: sign errors on the middle term, and picking the wrong 2×2 sub-matrix.

1.2 Function and Simple Examples

A function is a rule assigning exactly one output to each input, written f(x). If f: A → B, every element of A (domain) maps to exactly one element of B (range/codomain). One input, one output — if an input can give two different outputs, it is not a function.

Domain = all valid inputs. **Range** = all outputs actually produced.

Example 3

f(x) = x^2 + 3. Find f(2), f(–1), f(0).

f(2) = 4+3 = 7 f(–1) = 1+3 = 4 f(0) = 0+3 = 3

Example 4 — Domain check

f(x) = 1 / (x – 2). Denominator can't be zero → x ≠ 2. Domain = all reals except 2.

Example 5 — Domain check

f(x) = √(x – 5). Expression under the root must be ≥ 0 → x ≥ 5.

Where marks are lost: forgetting the denominator-can't-be-zero restriction, and forgetting that square roots need a non-negative argument. Check both, every time.

1.3 Logarithm as a Function

Logarithm answers: “to what power do I raise the base to get this number?”

$$\log_a N = x \iff a^x = N$$

a = base, N = number, x = the exponent (the answer). Conditions: a > 0, a ≠ 1, N > 0 — always. log of zero or a negative number does not exist.

Example 6

$\log_2 8 = 3$, because $2^3 = 8$.

Example 7

$\log_3 81 = 4$, because $3^4 = 81$.

Two types used constantly: Common log — $\log_{10} N$, written log N. Natural log — $\log_e N$, written ln N, where e ≈ 2.718.

As a function, $\log_a x$ takes any positive real number as input and gives a real number as output — domain x > 0, range = all reals. It is the mirror image of the exponential function a^x.

1.4 Laws of Logarithm and Related Simple Examples

These four laws are the entire game — every log problem in this course reduces to applying one or more of them.

Law	Statement
Product	$\log_a(MN) = \log_a M + \log_a N$
Quotient	$\log_a(M/N) = \log_a M - \log_a N$
Power	$\log_a(M^n) = n \log_a M$

Change of base	log■ N = log_b N / log_b a

Plus two identities to know instantly: log■ 1 = 0 (any base), and log■ a = 1.

Example 8 — Product law

Expand log(3 × 7): = log 3 + log 7

Example 9 — Quotient law

Expand log(25/4): = log 25 – log 4

Example 10 — Power law

Simplify log(x^3): = 3 log x

Example 11 — Combined (typical exam question)

Simplify log(a^2b / c^3):

= log(a^2b) – log(c^3) = log a^2 + log b – log c^3 = **2 log a + log b – 3 log c**

Example 12 — Solving an equation

Solve log■(x) = 5: x = 2■ = **32**

Example 13 — Change of base

Find log■ 8 using base 10: log■ 8 = log 8 / log 2 = 0.903 / 0.301 = **3**

Where marks are lost: treating log(M+N) as log M + log N — this is flat wrong. The product law is for log(MN), not log(M+N). There is no law for the log of a sum. Also: mixing up which term subtracts in the quotient law.

Self-Check Before Unit 2

You should be able to, without hesitation:

1. Evaluate a 3×3 determinant with correct signs
2. State the domain of a rational or square-root function
3. Convert between exponential form and log form
4. Apply all four log laws to simplify an expression in one pass

If any of these feel shaky, redo the relevant examples before Unit 2 — Trigonometry assumes fluency with this manipulation.